Solinas: point coverage and useful relations

Measured 2026-10-11 UTC • EXP-ECDLP-4c6a09 • H-ECDLP-4da36f

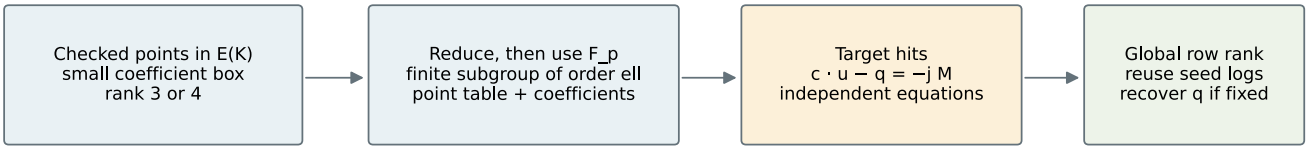
Added rank expands point supply; these linear-combination methods do not beat generic BSGS. With generator information shared, Solinas recovered all **96/96** public synthetic targets. BSGS recovered the same targets with less group work in every one of six matched curve/prime cohorts. These are observations, not independently reviewed claims, deployed-curve results or asymptotic bounds.

Largest prime: p = 68,719,472,639

Coefficients in [−4,4]; rational ranks 1 → 3 and 2 → 4 over the Solinas cubic field.

Curve rhs	Subgroup ell	Q → K points	K calls	BSGS calls	K/BSGS
x^3-3x+3	5,385,547	9 → 729	92,149	20,396	4.52×
x^3+7x+1	26,109,179	81 → 6,561	81,266	36,989	2.20×

K points cover 0.01354% and 0.02513% of these subgroups. Both are **81×** the rational-only point counts, with 81× as many coefficient vectors. Each row recovered 16/16 targets for both methods. Calls include seed projection, table construction and shared queries; exact-K and prior rank-discovery work are separate.



A collision counts as useful only when it adds modular matrix rank.
Known-scalar BSGS points avoid unknown seed logs.

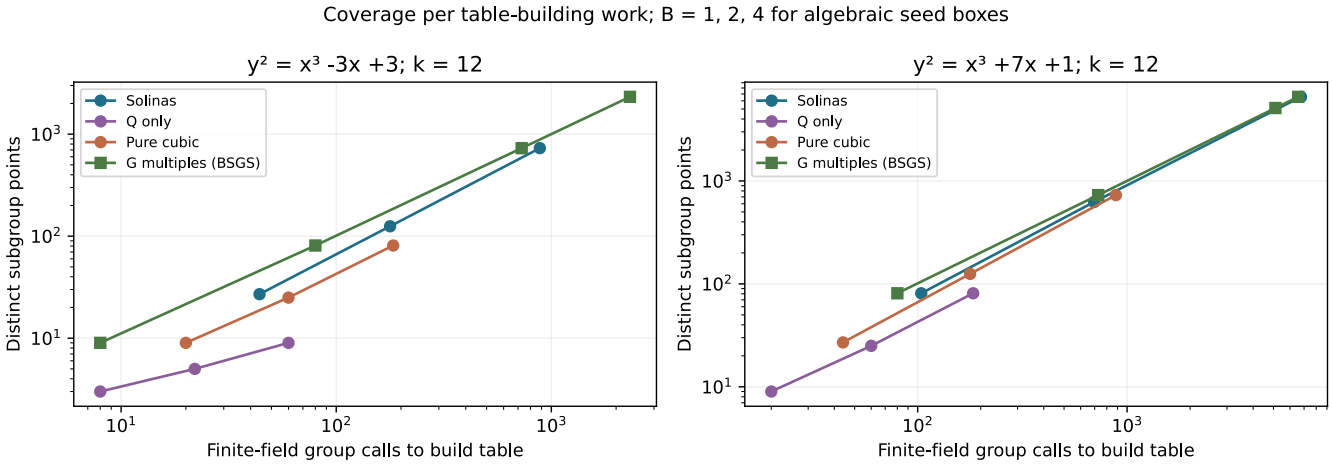
Measured toy pipeline; the field identity alone does not reveal u or q.

A useful relation is new information

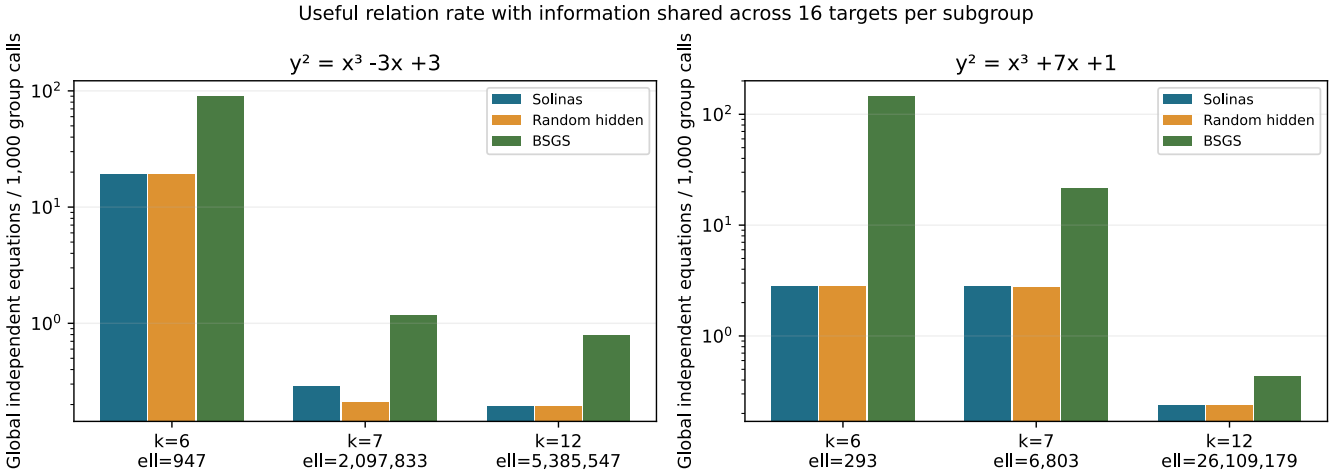
Unknown seed logs u and target logs q enter one modular linear system over F_ell. Collisions give (c−d)·u=0; a target hit gives c·u−q=−jM. Only matrix-rank increases count. A hit alone is not scalar recovery. Learned seed constraints are shared across all 16 targets, and q is returned only when uniquely implied.

At this prime, Solinas yields **0.1953** and **0.2338 independent relations per 1,000 group calls**; the respective BSGS controls yield **0.7845** and **0.4326**. The extra two or three equations pay for unknown seed logs. All recovered scalars and independent relation witnesses were checked.

Coverage and relation efficiency



Measured build operations at k=12. Algebraic boxes use B=1,2,4. Ordinary known multiples of G reach at least as many points per construction call. Seed projection and prior descent cost are excluded from this build-only chart.

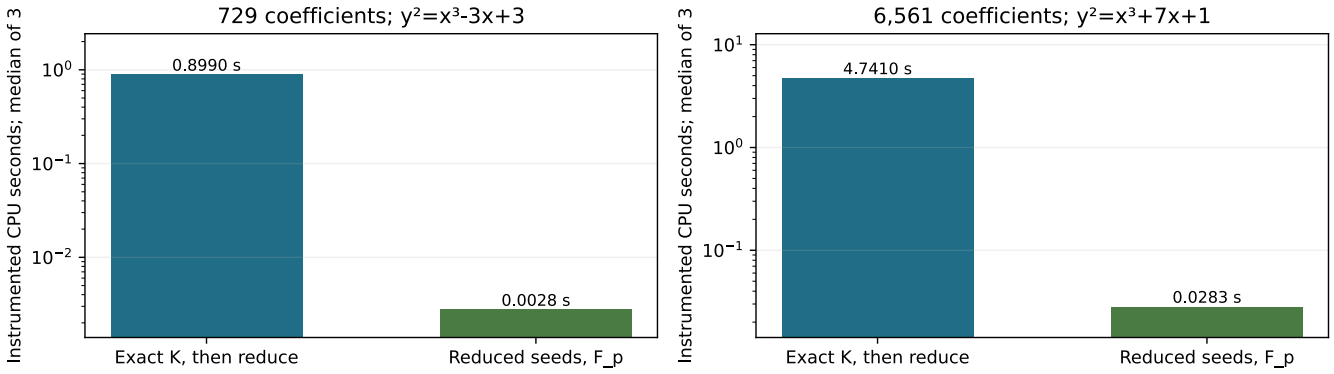


Six fixed cohorts, 16 common targets each; one random finite-field seed set per configuration. No population confidence interval is claimed. Random hidden seeds have matching seed count, known generator slot, box and target inputs. At k=12, Solinas takes 92,149 and 81,266 calls; the matching random hidden sets take 92,465 and 80,996. This run does not establish a special Solinas advantage.

BSGS denotes the best measured control among the predeclared matched table sizes and ceil(sqrt(ell)); it is not a claim of globally optimal batch parameters.

Exact field cost and matched controls

K=Q(alpha), alpha³−alpha−1=0, reduces through alpha → 2^k. All exact-K table reductions agree entry-for-entry with generation directly from reduced seeds. Once those seeds are known, the cheaper finite-field path gives the same points.



Three timing repeats. Component medians for exact-K precomputation, generation and reduction at k=12 total about **0.899 s** and **4.741 s**. Coordinate numerator/ denominator sizes reach 495 and 646 bits; this is not canonical height. GP timing includes bit-size checks and first-repeat row output; cached Python timing includes relation construction/checks. These machine timings are not portable primitive weights.

The preserved cold-target boundary resets learned generator constraints between targets: 14/16 and 16/16 recoveries at k=12, costing 261,072 and 230,529 calls. Its per-target relation counts are not global batch rank. The shared phase uses the identical point sets, target order and 32,768-probe cap, with one common query jump.

Controls and actual scope

- Two selected rank-gain models; three primes k=6,7,12; 78 coverage cells; 90 configurations in each cold/ shared phase; 96 unique synthetic targets.
- Pure cubic control y³−3: ranks 2 and 3. Random low-weight control y³+y−1: rank 1 and bounds [2,3]. The supplied point sets are not claimed saturated bases.
- The random field has no degree-one reduction at k=7,12. Four unavailable model/prime comparisons are retained; no substitute prime was selected.
- Sign-change control alpha=−beta, beta³−beta+1=0 matches all six Solinas cohorts. It is an isomorphic null, not an independent sample.
- A first GP preparation failed on t_FFELT conversion. The retained retry uses the identity trace in degree one. All failures and source versions are preserved.

Incomplete searches remain censored. Small subgroups, selected models and one random seed set restrict inference. Higher fields and deployed NIST curves are untested. Cheaper new relations or a better height-bounded lift mechanism remain separate opportunities; these measurements do not rule them out.

Evidence, identities and reproduction

Full identities bind the exact prime field, rational model, subgroup, cofactor and public generator. The same identity is used by every compared method. EC1P18Csolinaslifth8344f218e663 • A=-3, B=3, k=6
urn:ec-record:1:sha256:8344f218e66391b3a2c10bf4d3f538d257cd9b958603aa387e6fa9b017669b6c

EC1P21Csolinaslifth7f5aebdfe3ce • A=-3, B=3, k=7
urn:ec-record:1:sha256:7f5aebdfe3ce06dc6fb333edcf906eec91e660ae5e5a5f74411435f6f96635d1

EC1P36Csolinaslifth55d93546292c • A=-3, B=3, k=12
urn:ec-record:1:sha256:55d93546292c07c17b3387818426a571b1590bb818b7744fc040c477068d85b3

EC1P18Csolinaslifthaecb5e3fd711 • A=7, B=1, k=6
urn:ec-record:1:sha256:aecb5e3fd7115bdbfd214d14535a7fa414643bc3c2c9b3681d66d49a5ae7cac7

EC1P21Csolinaslifth2f53a922e551 • A=7, B=1, k=7
urn:ec-record:1:sha256:2f53a922e551928386c48f1afe98a9e2e9924dd67a49ef874908bcdc580b15ef

EC1P36Csolinaslifthc3394223ab66 • A=7, B=1, k=12
urn:ec-record:1:sha256:c3394223ab666ffa15217079258f3a281d9cacf2964ede7b08388d87b18b6985

Immutable sources

Run directory: experiments/EXP-ECDLP-4c6a09/runs/RUN-ECDLP-6d910b. Raw run commit: 2ee51a2b6a4c9919fd7102d8baae524d8cab80ec.

Principal result: benchmark-shared-attempt-01/raw-result.json. Frozen plan and shared-relations-plan.json define the boundaries; benchmark-attempt-01 retains the cold result. point-preparation-attempt-02/exact-grids retains exact GP tables and repeats. Earlier failed preparation and four matched descents remain beside them. Every recovered scalar was checked against both the public point and separately held synthetic truth; solver inputs exclude that truth.

Adjacent report.md links all evidence. methods.csv, paired-results.csv, coverage.csv and curve-identities.csv provide the full measurements. Editable build_report.py generates the SVG figures and PDF. Source snapshots, command arguments, logs, GPL-2 license, script hashes and integrity checks are retained. PARI/GP 2.17.2; Python 3.12.14. Reproduction uses fresh output locations.

No independent review, research-status change or sealed frontier-bound claim occurred. Historical coverage/rank reports remain unchanged; this new report adds the scoped observations. Canonical ECDLP frontier graphs require reviewed bound evidence and are unaffected. GitHub publication remains unavailable under the prior unchanged authentication failure; records are committed locally.